


# Bitcoin Performance Pipeline

An automated ELT pipeline and dashboard for BTC/USD, built on Databricks

**BINOVATE** Data engineering portfolio project

November 2026

THE BRIEF

# Binovate trades Bitcoin and wants its market view automated, not assembled by hand each morning.

### Automated

Data refreshes and the dashboard rebuilds on a daily schedule, with no manual steps.

### Low admin

One cloud platform, serverless compute, and checks that catch bad data on their own.

### Decision ready

Clean daily and weekly tables feed both a dashboard and a plain-English AI agent.

ARCHITECTURE

# A medallion ELT flow, source to dashboard



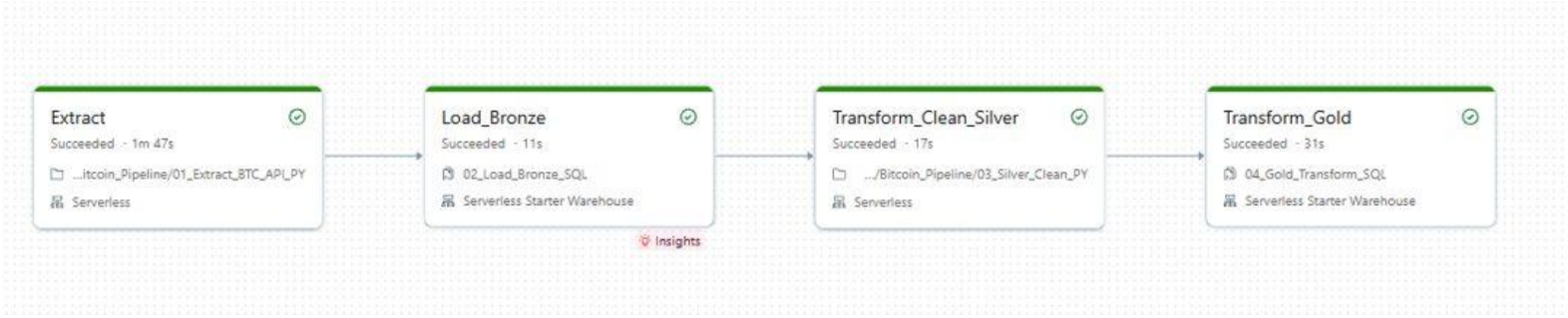

ELT, not ETL. Raw data lands first, then every transform runs inside Databricks.

Serverless notebooks and SQL, Delta tables, Parquet in a managed volume.

PIPELINE RUN

# Four chained tasks, one daily job

**2m 46s**
end to end, serverless




1 · Extract
**1m 47s**

2 · Load Bronze
**11s**

3 · Clean Silver
**17s**

4 · Transform Gold
**31s**

DATA QUALITY

# Bad bars are quarantined, not trusted

| 8,825 | 8,822 | 3 | 0 |
|---|---|---|---|
| bronze rows ingested | clean silver rows | bars quarantined | gaps in the 4h series |

## EVERY BAR IS CHECKED FOR

- Null timestamp, price or volume
- Price zero or negative
- Negative volume
- Zero volume or zero trades
- High below the open, close or low
- Low above the open, close or high
- Timestamp off the 4-hour grid
- Breaks in the 4-hour sequence

INSIGHT · PRICE

# BTC climbed toward $85k, then rolled over




**$84.9k**
peak weekly close, late September

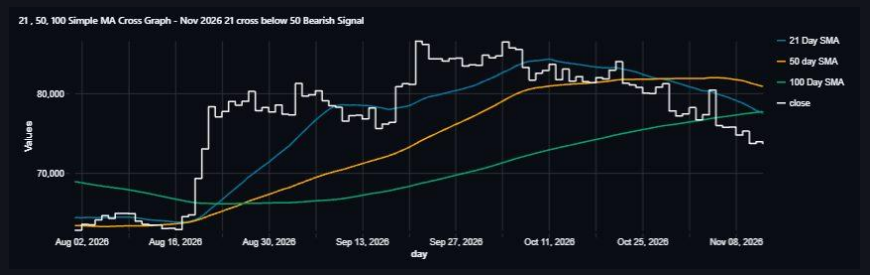
**$73.7k**
latest daily close, 12 Nov

**-5.9%**
over the last two weeks

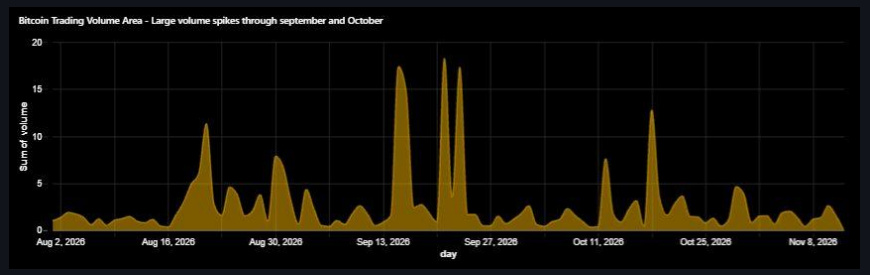
A steady summer climb from roughly $63k reversed into a sell-off, which the dashboard labels as bearish sentiment following a cyber attack.

INSIGHT · MOMENTUM

# Momentum has turned bearish




The 21-day average crossed below the 50-day in November, a classic bearish cross.




Volume spiked through September and October as the move accelerated.

**6 bearish weeks in a row**

Weekly change ran negative from early October after a run of green weeks over the summer.

AI AGENT

# Ask the data in plain English

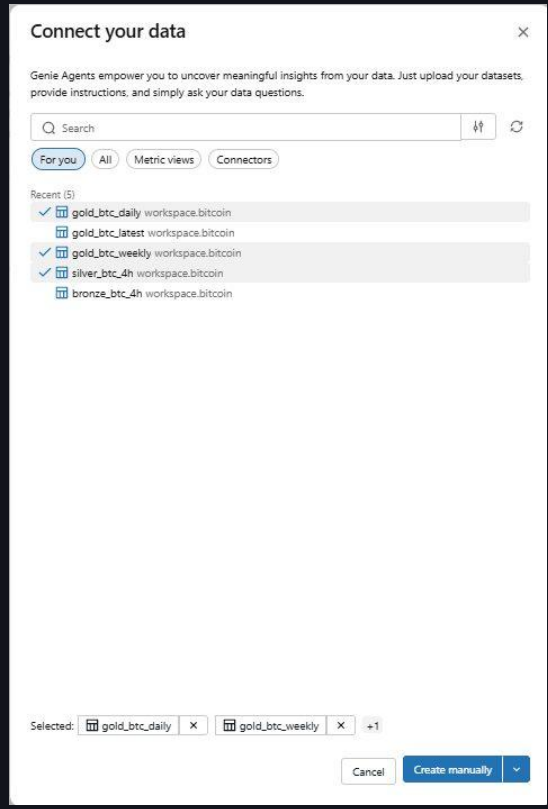
A Databricks Genie agent is wired to the Silver and Gold tables, so questions get answered without writing SQL.

“What is the latest Bitcoin price?”

“How did BTC move this week?”

“Is momentum bullish or bearish?”

- Reads one table per question: 4-hour, daily or weekly
- Percent in points, volume in BTC, all times UTC
- Skips incomplete periods, newest bar for the live price
- Frames every answer as history, never financial advice



OUTCOME

# A hands-off Bitcoin desk for Binovate

### Refreshes daily

New bars, clean tables and a rebuilt dashboard every day, unattended.

### Data you can trust

Eight validation checks and a quarantine table keep bad bars out.

### Answers on demand

A dashboard for the eye and an agent for the question.

NEXT

Add Commitment of Traders data · move notebooks to tested Python · widen the agent

Educational portfolio project. Nothing here is financial advice.